Clinical Handbook

Natural Language Edition from Markdown Source

Prepared for medical student learning and clinical reading

Introduction

This guideline consolidates and interprets clinical decision pathways from a structured case-based workbook focused on per vaginal bleeding (PVB) across diverse presentations. Each section corresponds to one worksheet in the source Excel file and represents a distinct clinical scenario — differentiated by dominant pathophysiology, gestational context, hemodynamic status, laboratory profile, or diagnostic certainty. Management is explicitly stratified by presentation: stable versus unstable, pregnant versus non-pregnant, septic versus non-septic, coagulopathic versus non-coagulopathic. All abbreviations are expanded at first use. Where local protocols differ, this document serves as a foundational reference — not a substitute for institutional policy or clinical judgment.

Condition 1: Hypovolemic Shock Secondary to Abnormal Uterine Bleeding

Clinical Presentation The patient presents with signs and symptoms consistent with acute hypovolemia due to excessive uterine blood loss. Common features include palpitations, dizziness, confusion, orthostatic hypotension (a drop in systolic blood pressure ≥20 millimetres of mercury upon standing), tachycardia (>100 beats per minute), delayed capillary refill (>3 seconds), and reduced urine output (<30 millilitres/hour). There is no evidence of external genital trauma or local vaginal lesion on speculum examination. Key Investigations Full blood count (complete blood count), including hemoglobin concentration, hematocrit, white blood cell count, and platelet count Coagulation profile: prothrombin time (PT), activated partial thromboplastin time (activated partial thromboplastin time), fibrinogen level Blood group and Rhesus status, and crossmatch for at least two units of packed red blood cells Serum electrolytes, urea, and creatinine Point-of-care lactate measurement Pelvic ultrasound to assess endometrial thickness, presence of retained products, or structural abnormalities Diagnostic Considerations Abnormal uterine bleeding (AUB) is defined as bleeding from the uterine corpus that is abnormal in volume, frequency, regularity, or duration, and occurs in the absence of pregnancy or identifiable pelvic pathology. In this scenario, AUB has progressed to hypovolemic shock — a life-threatening emergency requiring immediate intervention. Differential diagnoses include anovulatory bleeding, endometrial hyperplasia, submucosal fibroids, and malignancy; however, stabilization takes absolute priority over definitive diagnosis. Management Pathway Management follows a tiered, physiology-directed approach include Immediate resuscitation: Secure intravenous access with two large-bore (14–16 gauge) cannulas. Administer isotonic crystalloid (0.9% sodium chloride) at 1–2 liters rapidly, reassessing response after each bolus. If hemoglobin is <7 grams per decilitre or the patient remains unstable despite fluid, transfuse packed red blood cells. Pharmacological control of bleeding: Intravenous tranexamic acid (1 gram over 10 minutes, repeatable once after 30 minutes if bleeding persists) is indicated within 3 hours of onset. Antibiotic prophylaxis: Intravenous ceftriaxone (2 grams) plus metronidazole (500 mg) is administered to prevent infection, particularly if instrumentation or surgical intervention is anticipated. Gynecological assessment: Urgent referral to gynecology for evaluation of bleeding source, consideration of medical therapy (e.g., high-dose estrogen-progestin or progesterone-only regimens), and procedural options (e.g., endometrial sampling, hysteroscopy, or uterine artery embolization). Supportive Care Continuous monitoring of vital signs, urine output, and mental status Oxygen supplementation if oxygen saturation falls below 94% on room air Correction of electrolyte abnormalities (e.g., potassium, magnesium) Avoidance of nonsteroidal anti-inflammatory drugs (nonsteroidal anti-inflammatory drugs) until bleeding is controlled Admission Criteria All patients diagnosed with hypovolemic shock secondary to abnormal uterine bleeding require admission to an acute gynecology or high-dependency unit for ongoing observation, repeat hemoglobin monitoring, and multidisciplinary management.

Condition 2: Septic Shock Complicating Incomplete Abortion

Clinical Presentation The patient presents with fever (temperature ≥38.0°C), chills, lower abdominal pain, foul-smelling vaginal discharge, and signs of systemic inflammatory response — including tachycardia, tachypnea, and altered mental status. On examination, the uterus is enlarged, tender, and boggy. Cervical motion tenderness may be present. There is often active vaginal bleeding mixed with necrotic tissue. Key Investigations Complete blood count with differential (frequently showing leukocytosis or left shift, and sometimes granulocytosis or thrombocytopenia) C-reactive protein (C reactive protein) and procalcitonin levels Blood cultures (before antibiotics, if feasible) High vaginal and endocervical swabs for culture and sensitivity Pelvic ultrasound to confirm retained products of conception, assess for intrauterine gas or abscess formation, and rule out ectopic pregnancy Serum lactate, arterial blood gas, and renal function tests Diagnostic Considerations Incomplete abortion — defined as partial expulsion of pregnancy tissue with continued bleeding and risk of infection — is the underlying event. When accompanied by systemic signs of infection and organ dysfunction (e.g., acute kidney injury, altered consciousness, or respiratory distress), the diagnosis escalates to septic shock. This represents a critical deterioration requiring combined obstetric, infectious disease, and intensive care input. Management Pathway Management integrates sepsis bundle principles with urgent obstetric intervention include Sepsis resuscitation: Administer broad-spectrum intravenous antibiotics within 1 hour of recognition: ceftriaxone (2 grams IV) plus metronidazole (500 mg IV) is recommended as first-line. Add gentamicin (5 milligrams per kilogram IV) if Gram-negative sepsis is strongly suspected. Source control: Surgical evacuation of the uterus (manual vacuum aspiration or dilation and curettage) is performed as soon as feasible — ideally within 6 hours of diagnosis — to remove infected and necrotic tissue. Antibiotics should not delay evacuation. Hemodynamic support: Intravenous fluid resuscitation (crystalloid) guided by dynamic parameters (e.g., stroke volume variation, passive leg raise response); vasopressors (e.g., norepinephrine) initiated if persistent hypotension despite 30 millilitres/kg fluids. Adjunctive measures: Intravenous tranexamic acid is contraindicated in active sepsis unless life-threatening hemorrhage is present and outweighs infection risk. Supportive Care Strict aseptic technique during all procedures Serial lactate measurements every 2–4 hours until trending downward Monitoring for disseminated intravascular coagulation (disseminated intravascular coagulation): repeated fibrinogen, D-dimer, and platelet counts Psychological support and counseling regarding pregnancy loss and future fertility Admission Criteria Admission to a high-dependency or intensive care unit is mandatory. Patients must remain under continuous observation until hemodynamically stable for at least 24 hours, with C reactive protein trending downward and no new organ dysfunction.

Condition 3: Coagulopathy-Associated Uterine Hemorrhage

Clinical Presentation The patient presents with prolonged or recurrent vaginal bleeding without obvious structural cause, often with additional bleeding manifestations — such as easy bruising, epistaxis, gingival bleeding, or prolonged bleeding after minor cuts or procedures. There may be no hemodynamic instability initially, but bleeding can escalate rapidly. History may reveal known hematologic disorders (e.g., von Willebrand disease, immune thrombocytopenia), recent anticoagulant use (e.g., warfarin, direct oral anticoagulants), or recent chemotherapy. Key Investigations Full blood count with peripheral blood smear Coagulation screen: prothrombin time (PT), activated partial thromboplastin time (activated partial thromboplastin time), international normalized ratio (international normalized ratio), fibrinogen, and D-dimer Platelet function assay or von Willebrand factor antigen and activity testing, if clinically indicated Drug levels (e.g., apixaban or rivaroxaban anti-Xa activity) if direct oral anticoagulant use is suspected Liver function tests and renal function tests Diagnostic Considerations Uterine hemorrhage in the setting of coagulopathy reflects failure of normal hemostasis rather than anatomical or hormonal pathology. The diagnosis requires integration of clinical history, bleeding pattern, physical findings, and targeted laboratory testing. Acquired coagulopathies (e.g., due to liver disease, vitamin K deficiency, or anticoagulant overdose) are more common than inherited ones in acute settings. Management Pathway Management prioritizes correction of the underlying coagulopathy while controlling uterine bleeding include Reversal of anticoagulation: For warfarin-induced over-anticoagulation (international normalized ratio >5.0 with bleeding), administer intravenous vitamin K (10 mg) plus four-factor prothrombin complex concentrate (prothrombin complex concentrate). For direct oral anticoagulants, specific reversal agents (e.g., andexanet alfa for factor Xa inhibitors) are preferred when available; otherwise, prothrombin complex concentrate may be considered. Platelet support: Platelet transfusion is indicated if platelet count is <50 × 10■/L and active bleeding is present. Fibrinogen replacement: Cryoprecipitate or fibrinogen concentrate is administered if fibrinogen is <1.5 g/L. Tranexamic acid: Intravenous tranexamic acid (1 gram over 10 minutes) remains safe and effective, even in many coagulopathic states, and should be given promptly. Definitive gynecological management: Once coagulation is optimized, consider hormonal therapy (e.g., high-dose intravenous conjugated equine estrogens) or surgical options — but only after correction of coagulopathy to avoid catastrophic hemorrhage. Supportive Care Avoid intramuscular injections and invasive procedures until coagulation parameters normalize Monitor for thrombotic complications following coagulation factor replacement Referral to hematology for long-term diagnosis and management of underlying disorder Admission Criteria All patients require inpatient admission for close hemostatic monitoring, repeat coagulation studies, and coordinated multidisciplinary care involving gynecology, hematology, and pharmacy.

Condition 4: Pregnancy-Related Bleeding with Fetal Viability Concerns

Clinical Presentation The patient is pregnant and presents with vaginal bleeding — ranging from light spotting to heavy hemorrhage — with or without associated abdominal pain or uterine contractions. Vital signs may be stable, but concern centers on both maternal safety and fetal well-being. Symptoms such as decreased fetal movement, leakage of amniotic fluid, or uterine rigidity suggest possible placental abruption or preterm labor. Key Investigations Fetal heart rate monitoring (continuous or intermittent, depending on stability) Transvaginal or transabdominal pelvic ultrasound to determine gestational age, fetal viability, placental location and appearance, presence of retroplacental clot, and amniotic fluid volume Maternal complete blood count, blood group and Rhesus status, and Rhesusesus antibody screen Cervical length measurement if preterm presentation is suspected Uterine activity monitoring if contractions are reported Diagnostic Considerations Bleeding in pregnancy carries multiple etiologies: threatened miscarriage, inevitable or incomplete abortion, ectopic pregnancy, cervical polyp or ectropion, placenta previa, placental abruption, or vasa previa. Accurate diagnosis depends on integrating gestational age, ultrasound findings, and clinical trajectory. Fetal heart rate patterns provide crucial insight into fetal compromise — late decelerations, minimal variability, or bradycardia signal acute hypoxia. Management Pathway Management is gestational-age-dependent and risk-stratified include Before 24 weeks: Focus shifts to maternal stabilization and diagnosis. If ectopic pregnancy is suspected, serum beta-human chorionic gonadotropin (β-hCG) serial measurement and early ultrasound guide management. If viable intrauterine pregnancy is confirmed and bleeding is light without pain, expectant management with serial ultrasound may be appropriate. At or beyond 24 weeks: Fetal surveillance becomes central. Persistent non-reassuring fetal heart rate patterns or evidence of placental abruption mandate delivery — by induction of labor or cesarean delivery, depending on urgency and cervical readiness. Rhesus-negative mothers: Administer anti-Rhesus(D) immunoglobulin

(300 micrograms intramuscularly) within 72 hours of any episode of vaginal bleeding, regardless of gestational age or volume. Pharmacological support: Tocolytics are contraindicated in the presence of active bleeding or placental abruption. Corticosteroids for fetal lung maturity are indicated if delivery is anticipated between 24 and 34 weeks. Supportive Care Counseling on pregnancy prognosis and psychosocial impact Access to bereavement support services if pregnancy loss occurs Clear documentation of fetal assessments and maternal decisions Admission Criteria All pregnant patients presenting with vaginal bleeding require admission for at least 24 hours of observation, fetal surveillance, and serial assessment — especially if gestation is ≥24 weeks, bleeding is moderate-to-heavy, or fetal heart rate abnormalities are detected.

Condition 5: Postpartum Hemorrhage Beyond the Immediate Puerperium

Clinical Presentation The patient presents with vaginal bleeding occurring between 24 hours and 12 weeks after delivery — classified as secondary postpartum hemorrhage. Bleeding may be sudden and heavy, often accompanied by clots, uterine tenderness, and low-grade fever. On examination, the uterus may be enlarged and boggy, with a partially open cervical os and retained tissue palpable through the cervix. Key Investigations Complete blood count and coagulation profile High vaginal swab for aerobic and anaerobic culture Pelvic ultrasound to identify retained products of conception, uterine cavity fluid, or focal myometrial abnormalities C-reactive protein and erythrocyte sedimentation rate Endometrial sampling (if malignancy or chronic endometritis is suspected) Diagnostic Considerations Secondary postpartum hemorrhage is most commonly caused by retained products of conception or endometritis. Less common causes include uterine artery pseudoaneurysm, arteriovenous malformation, or gestational trophoblastic disease. Differentiation between infectious and non-infectious etiologies guides antibiotic use and timing of intervention. Management Pathway Management integrates antimicrobial, medical, and procedural strategies include Empirical antibiotics: Intravenous ceftriaxone (2 grams) plus metronidazole (500 mg) is initiated immediately if fever or uterine tenderness is present, pending culture results. Medical evacuation: Misoprostol (800 micrograms vaginally) or mifepristone (200 mg orally) may be trialed in stable patients with small-volume retained tissue and no signs of sepsis. Surgical evacuation: Manual removal of retained products under ultrasound guidance or dilation and curettage is indicated for heavy bleeding, hemodynamic instability, or failure of medical management. Tranexamic acid: Intravenous tranexamic acid (1 gram) is administered concurrently with evacuation to reduce blood loss. Uterotonics: Intramuscular or intravenous oxytocin (10 units) or carboprost (250 micrograms) may be used if uterine atony contributes to bleeding. Supportive Care Monitoring for signs of pelvic abscess or septic shock Follow-up ultrasound 48–72 hours after evacuation to confirm cavity clearance Counseling on contraception and future pregnancy planning Admission Criteria All patients require admission for at least 48 hours post-intervention for observation, repeat hemoglobin checks, and assessment of response to antibiotics or surgery.

Appendix: Principles of Hypovolemic Shock Resuscitation

These principles underpin the management of all hemorrhagic conditions described above include Goal-directed fluid therapy: Use dynamic markers (e.g., stroke volume variation, pulse pressure variation) over static markers (e.g., central venous pressure) where available. Target mean arterial pressure ≥65 millimetres of mercury, urine output ≥0.5 millilitres/kilogram per hour, and normalization of lactate. Blood product ratios in massive transfusion: For patients requiring >10 units of packed red blood cells in 24 hours, follow a 1:1:1 ratio of packed red blood cells: fresh frozen plasma: platelets — adjusted by viscoelastic testing (e.g., thromboelastography) when available. Avoid routine use of colloids: Crystalloids remain first-line. Albumin may be considered in specific hypoalbuminemic states, but hydroxyethyl starches are contraindicated due to renal and coagulation risks. Early involvement of senior clinicians: Escalate to consultant-led care when any of the following occur: transfusion of >4 units of blood, persistent lactate >4 mmol/L after 2 hours, development of coagulopathy, or need for intensive care unit-level monitoring.

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